

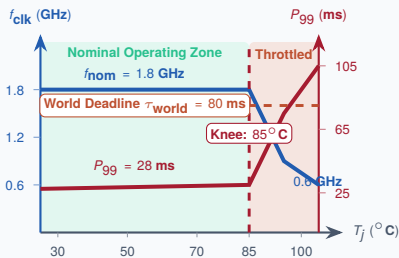


THERMAL DERATING CLIFF & THE RE-PLACEMENT RIPPLE EFFECT

Quantifying How Silicon Junction Temperature and Workload Re-Placement Mutate Latency, Power, and Stopping Physics

(A) THE THERMAL DERATING KNEE ($T_j > 85^\circ\text{C}$)

Dynamic Voltage & Frequency Scaling (DVFS) Inflates P_{99} Latency Tail



Physical Systems Consequence:

When $T_j > 85^\circ\text{C}$, DVFS throttles clock from 1.8 GHz $\rightarrow$ 600 MHz. Inference latency triples ($P_{99} = 105\text{ ms} > \tau_{\text{world}}$), inflating physical stopping distance d_{stop} from 0.35 m $\rightarrow$ 0.82 m, breaching safety clearance.

(B) RE-PLACEMENT RIPPLE: CLOUD $\leftrightarrow$ EDGE NPU

Multi-Dimensional Tradeoffs When Relocating the VLA Policy

Cloud vs. Edge NPU Placement Comparison:

Systems Dimension	Cloud 70B VLA	Edge 3B INT4 NPU
Model Parameter Scale	70 Billion FP16	3 Billion INT4
Inference Tail (P_{99})	480 ms (High Jitter)	35 ms (Deterministic)
Wireless Dependency	100% (5G MEC Required)	0% (Local Autonomous)
On-Board Thermal (TDP)	0 Watts	+8.5 Watts Heat Load
Chassis Battery Life	4.2 Hours	3.1 Hours (-26%)
Stopping Envelope (d_{stop})	0.85 m Margin	0.32 m Margin (-62%)
Safe Max Transit Speed	0.6 m/s	1.5 m/s ($+150\%$)

The Placement Ripple Theorem:

Moving a neural model from Cloud to Edge shrinks stopping distance by 62% and eliminates network disconnect hazard, but consumes +8.5 W of battery power and demands dedicated heatsink airflow.

Engineering Rule of Thumb:

If $d_{\text{stop}}(\text{Cloud}) > d_{\text{clearance, min}}$, local Edge NPU placement is non-negotiable regardless of on-board battery and thermal costs.